

MODULE 04 · DOMAIN 3 - APPLICATIONS OF FOUNDATION MODELS

Amazon Bedrock in Depth

Choosing a model, controlling inference, and paying for it sensibly.

DOMAIN 3 of 5 EXAM WEIGHT 28% MODULE 4 of 14

Bedrock is **one API** in front of many foundation models. Serverless, so there is **no infrastructure to manage.**

The parts of Bedrock worth naming

- ◆ **Model access** — Amazon Nova and Titan, Anthropic, Meta, Mistral, Cohere, AI21, Stability.
- ◆ **Customisation** — fine-tuning, continued pre-training, distillation, custom model import.
- ◆ **Knowledge Bases** for RAG, **Agents** for tool use, **Flows** for orchestration.
- ◆ **Guardrails** for safety, **Prompt Management** for versioned prompts, **Evaluations** for comparison.
- ◆ **Your data stays yours.** It is not used to train the base models.

ON THE EXAM

That last point answers a recurring question about whether prompts sent to Bedrock train the underlying model. They do not.

What actually decides which model

CRITERION	WHAT TO WEIGH
Capability	Can it do the task at all? Reasoning depth, coding, long documents.
Modality	Text only, or does it need to accept images, video or audio?
Latency	Interactive chat needs a small fast model. Batch work does not.
Cost	Price per million tokens varies by roughly two orders of magnitude.
Context window	How much must fit in one request, prompt plus response.
Customisability	Not every model supports fine-tuning.

ON THE EXAM

Questions rarely ask which model is best. They describe a constraint and ask which model fits it. Find the constraint first.

The phrases that decide it

- ◆ 'Real-time', 'low latency', 'interactive' — smallest model that can do the job.
- ◆ 'Most cost-effective', 'high volume' — cheapest adequate model, and consider batch.
- ◆ 'Complex reasoning', 'nuanced' — a larger model, accepting the cost.
- ◆ 'Analyse images', 'read the chart' — multimodal input, so Nova Lite or Pro.
- ◆ Bigger is not better by default. Over-specifying is a wrong answer, not a safe one.

ON THE EXAM

The largest model is the correct answer only when the scenario names a hard task. Otherwise it is the expensive trap.

The dials you turn at request time

- ◆ **Temperature** — how much randomness in choosing the next token.
- ◆ **Top-p** — sample only from the smallest set of tokens whose probabilities sum to p .
- ◆ **Top-k** — sample only from the k most likely tokens.
- ◆ **Max tokens** — a ceiling on response length. Controls cost and cut-off.
- ◆ **Stop sequences** — strings that end generation when produced.

ON THE EXAM

None of these change the model. They change how the next token is chosen from what the model already produced.

Low temperature or high?

LOWER TEMPERATURE

Focused and repeatable

- Picks high-probability tokens
- Consistent answers across runs
- Factual Q and A, extraction, classification
- Reduces invention and drift

HIGHER TEMPERATURE

Varied and creative

- Gives unlikely tokens a real chance
- Different answer each time
- Brainstorming, marketing copy, story ideas
- Increases the risk of hallucination

THE TIEBREAKER

Accuracy and consistency mean lower. Creativity and variety mean higher. If a question says answers must be reliable, lower temperature is in the correct answer.

Two ways to shrink the candidate pool

- ◆ **Top-k** keeps a fixed number of candidates. Top 50 tokens, regardless of confidence.
- ◆ **Top-p** keeps a variable number — however many are needed to reach the probability mass.
- ◆ Top-p adapts: few candidates when the model is confident, more when it is unsure.
- ◆ Adjust temperature **or** top-p, not both at once. They interact confusingly.

ON THE EXAM

Both narrow which tokens can be chosen. Temperature reshapes the probabilities; top-p and top-k truncate the list.

Max tokens is a cost control, not a style control

- ◆ Max tokens truncates. It does not teach the model to be concise.
- ◆ Set it too low and answers are cut off mid-sentence.
- ◆ For genuinely shorter answers, ask for brevity **in the prompt**.
- ◆ Stop sequences end generation cleanly at a marker you define.

ON THE EXAM

A question about responses being cut off mid-sentence is a max tokens question. A question about responses being too verbose is a prompt question.

Three purchase modes, and that is the whole menu

MODE	HOW IT WORKS
On-demand	Pay per input and output token. No commitment. The default.
Batch	Submit a file of prompts, results land in S3. Roughly 50% cheaper.
Provisioned Throughput	Reserve dedicated capacity in model units, billed hourly.

ON THE EXAM

Batch is the answer whenever a scenario says results are not needed immediately and cost matters. It is the single most reusable cost fact in this domain.

On-demand or provisioned throughput?

ON-DEMAND

Pay for what you use

- No commitment, no minimum
- Variable or unpredictable traffic
- Development and early production
- Subject to shared rate limits

PROVISIONED THROUGHPUT

Reserve the capacity

- Hourly rate, committed for 1 or 6 months
- Steady, high, predictable volume
- Guaranteed throughput, no throttling
- Required to host custom models

THE TIEBREAKER

Steady high volume, or a guarantee against throttling, means provisioned. Spiky or unknown traffic means on-demand.

A fine-tuned or imported model on Bedrock can only be hosted with **provisioned throughput.**

Half price, if you can wait

- ◆ Submit many prompts as a file in S3. Results are written back to S3.
- ◆ Roughly **50% off** on-demand rates for supported models.
- ◆ Asynchronous. Typically completes within hours, not seconds.
- ◆ Useless for anything a user is waiting on.

ON THE EXAM

Nightly summarisation, bulk classification, dataset labelling and embedding backfills are all batch scenarios.

Two more ways the bill comes down

- ◆ **Prompt caching** — reuse a repeated prompt prefix instead of paying for it every call.
- ◆ Large savings when a long system prompt or document is sent on every request.
- ◆ **Model distillation** — a large teacher model trains a smaller, cheaper student.
- ◆ The student runs at a fraction of the cost with much of the quality.

ON THE EXAM

Distillation is the answer when a scenario wants the quality of a large model at the cost and latency of a small one.

Four ways to make a model yours

METHOD	WHEN IT APPLIES
Prompt engineering	Free and instant. Always try this first.
RAG	The model lacks facts. Grounds answers in your documents.
Fine-tuning	The model lacks behaviour. Needs labelled pairs.
Continued pre-training	The model lacks vocabulary. Needs unlabelled domain text.

ON THE EXAM

The exam expects this order. Cheapest and simplest first, and only escalate when the previous option cannot solve it.

Bringing your own weights

- ◆ Import a model you trained elsewhere, including on SageMaker, and serve it through Bedrock.
- ◆ You get the Bedrock API, Guardrails and tooling around your own model.
- ◆ Supported architectures only. Not any arbitrary model.
- ◆ Hosted on provisioned throughput, so the standing cost applies.

ON THE EXAM

This is the answer when a company already has a trained model but wants Bedrock's API and surrounding features.

Prompt Management, Flows and the playground

- ◆ **Prompt Management** — store prompts as versioned resources with variables.
- ◆ Update a prompt without redeploying the application. Compare versions before promoting.
- ◆ **Flows** — chain prompts, models, Knowledge Bases and Lambda into a visual workflow.
- ◆ **PartyRock** — a free, no-code playground for learning. Not a production tool.

ON THE EXAM

A team wanting to change prompt wording without a code deploy is a Prompt Management question.

Carry these into the exam

- ◆ Bedrock is one serverless API across many providers. Swapping models is a parameter change.
- ◆ Lower temperature for accuracy, higher for creativity. It is the most-tested parameter.
- ◆ Max tokens truncates. It does not make the model concise.
- ◆ Not needed immediately plus cost matters means batch, at about half price.
- ◆ Steady high volume or no-throttling means provisioned throughput.
- ◆ Fine-tuned and imported models can only be hosted on provisioned throughput.

A publisher runs a nightly job that summarises about eighty thousand news articles. The summaries are read by editors the following morning.

The current on-demand bill is higher than the team expected and they have been asked to reduce it.

What should they change first?

- ☐ A Buy provisioned throughput
- ☐ B Switch the job to batch inference
- ☐ C Lower the temperature
- ☐ D Fine-tune a smaller model

◆ PAUSE AND DECIDE

What should they change first?

- A

 Provisioned throughput
Suits steady all-day traffic. This job runs once a night.
- B

 Batch inference
Nobody is waiting until morning. Roughly half the cost.
- C

 Lower the temperature
Changes output style. Costs exactly the same.
- D

 Fine-tune a smaller model
Adds a training cost and a standing hosting bill.

ON THE EXAM

'Read the following morning' is the batch signal. The scenario tells you nobody is waiting.

A bank's compliance assistant answers policy questions for staff. Testing shows that asking the same question twice sometimes produces differently worded answers.

Compliance requires that identical questions receive consistent answers.

Which change best addresses this?

- ☐ A Lower the temperature
- ☐ B Raise max tokens
- ☐ C Switch to provisioned throughput
- ☐ D Increase top-k

◆ PAUSE AND DECIDE

Which change best addresses this?

- A

 Lower the temperature
Concentrates sampling on likely tokens. More consistent output.
- B

 Raise max tokens
Allows longer answers. Nothing to do with consistency.
- C

 Provisioned throughput
A capacity and billing choice. No effect on output.
- D

 Increase top-k
Widens the candidate pool. Makes variation worse.

ON THE EXAM

Note option D moves in the wrong direction. Increasing top-k admits more candidates, so output varies more, not less.

A company fine-tuned a foundation model on Bedrock for a specialised classification task. Training finished successfully.

The team assumed inference would be billed per token as before. Finance is now asking why there is a large recurring hourly charge.

What is the explanation?

- ☐ A Fine-tuned models are billed at a higher per-token rate
- ☐ B Custom models require provisioned throughput to be hosted
- ☐ C The training job is still running
- ☐ D Guardrails are charged hourly

◆ PAUSE AND DECIDE

What is the explanation?

- A

 Higher per-token rate
Plausible, but custom models are not served on-demand at all.
- B

 Provisioned throughput is required
Hourly capacity charge, running whether or not it is called.
- C

 Training still running
The scenario states training finished.
- D

 Guardrails charged hourly
Guardrails are usage-based, and none were mentioned.

ON THE EXAM

The hourly shape of the charge is the clue. Token billing produces a variable bill; reserved capacity produces a flat one.

A retailer sends the same four-page returns policy with every customer service request, followed by the customer's question.

Volume is high and steady. The team wants to cut cost without changing the model or the answers.

Which approach fits best?

- ☐ A Enable prompt caching
- ☐ B Reduce max tokens
- ☐ C Move to batch inference
- ☐ D Fine-tune the model on the policy

◆ PAUSE AND DECIDE

Which approach fits best?

- A** Prompt caching
The repeated prefix is paid for once, not on every call.
- B** Reduce max tokens
Truncates answers. The scenario forbids changing them.
- C** Batch inference
Customers are waiting. Batch is asynchronous.
- D** Fine-tune on the policy
Changes the model, adds hosting cost, and policies change.

ON THE EXAM

An identical prefix on every request is the prompt caching signature. Note that batch is wrong here purely because a customer is waiting.